Machine Learning Framework for Rental Investment Analysis in College Towns Across the U.S.

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Abstract

This practicum project focuses on creating a machine learning model to evaluate the rental investment potential across the U.S. The project explores the challenge of evaluating student rental markets, where housing demand is affected by multiple factors, such as rental prices, home values, university enrollment, demographic trends, and local economic conditions. Publicly available datasets from Zillow, the American Community Survey (ACS), and the Integrated Postsecondary Education Data System (IPEDS), and the Federal Reserve Economic Data (FRED) will be integrated to create a unified dataset for analysis.

The project will include data collection, cleaning, exploratory analysis, feature engineering, and predictive modeling. Machine learning models will be used to identify patterns associated with rental market conditions and investment potential. The main question is whether housing, demographic, and university data can be used to better predict rental investment opportunities across college-town markets.

I. Introduction/Background

College-town housing markets can be difficult to evaluate because rental demand is affected by several associated factors, including student enrollment, housing prices, rental rates, local income levels, and general economic conditions. In addition to the short-term financial risks, poor investment decisions can lead to long vacancies, unstable rental income, and difficulty adjusting to changing housing markets. Although large amounts of housing and demographic data are publicly available, these datasets are often stored separately, use different geographic structures, and can be difficult to analyze together over long periods.

Student rental housing is linked to university enrollment trends, local economic conditions, and demographic changes. Property owners and investors commonly rely on personal experience or general market patterns when making decisions. These approaches may omit important patterns or relationships that influence long-term rental market performance. Data science methods cannot replace market experience, but they may help identify trends and interactions that are difficult to consistently recognize through manual analysis alone.

While studying data science, many projects focus only on a single stage of the process, such as exploratory analysis or predictive modeling. However, real-world housing analysis requires the full data science pipeline, including data collection, cleaning, integration, feature engineering, modeling, validation, and result interpretation. This practicum focuses on applying that complete process to a real-world housing problem using publicly available datasets with different structures, geographic levels, and update frequencies.

The idea behind this practicum is that housing, demographic, and university data can be combined into a reproducible analytical framework able to evaluate rental investment potential in college towns across the U.S. Data from Zillow [1], the American Community Survey (ACS) [2], IPEDS [3], and FRED [4] will be integrated to analyze relationships between rental prices, home values, student enrollment, and demographic trends. Machine learning models will then be used to identify factors associated with stronger rental market conditions and investment opportunities.

The relevance of this practicum is that many real estate analytics projects stop at descriptive statistics or market summaries. Instead, this project moves beyond description toward predictive analysis by evaluating factors that may influence rental investment potential and comparing conditions across college-town markets. The project also aims to demonstrate how housing, demographic, economic, and university data can be integrated into a single analytical process that supports more informed, data-driven decision-making.

II. Problem Statement

Starting a college-town rental investment can be difficult because housing demand is influenced by multiple factors, including student enrollment, rental prices, home values, income levels, and local economic conditions. Many smaller property owners rely on personal experience or general market trends when making investment decisions, but these approaches may not fully capture how rental markets change over time or which factors most strongly influence investment potential.

There is also a large amount of publicly available housing and demographic data, but much of it exists in separate systems with different structures, geographic levels, and update frequencies. Housing data, demographic information, and university enrollment statistics are often analyzed separately, making it difficult to identify broader relationships between student populations, rental demand, and housing market conditions.

This project addresses the lack of a reproducible, data-driven framework that combines housing, demographic, and university data to evaluate rental investment potential in college-town markets. The project focuses on college towns across the U.S. and is partially motivated by personal experience managing a student rental property in Colorado. The main question guiding this project is: "What factors most strongly influence rental investment potential in Colorado college towns, and can machine learning models be used to predict stronger investment opportunities?"

To address this question, the project will use the following stages of the data science process: data collection, cleaning, integration, exploratory analysis, feature engineering, predictive modeling, and reporting. Data from Zillow, ACS, IPEDS and FRED will be combined to analyze relationships between rental prices, home values, student enrollment, and demographic trends.

III. Related Work

Research on housing analytics and real estate prediction has increased alongside the growth of publicly available housing datasets and advances in machine learning. Earlier studies in real estate analysis relied on traditional statistical methods and on structured housing variables such as property size, number of rooms, and specific neighborhoods. Recent work utilizes machine learning, geospatial analysis, and large housing datasets to improve predictive modeling and support decision-making in housing markets.

Researchers studied how to predict rental prices using machine learning. Yoshida et al. (2024) compared regression and machine learning methods for apartment rents with spatial data and found that machine learning performed better with larger datasets [5]. Maia and David (2024) applied machine learning to rental pricing in São Paulo and found that adding geographic information to housing features improved prediction accuracy in urban markets [6]. Baur et al. (2023) found that including property descriptions in machine learning models reduced prediction errors and improved automated real estate valuation [7].

Recent research has compared different machine learning methods for predicting rental prices. Kate et al. (2025) developed a framework for predicting house rentals and making property recommendations, showing that these models can benefit both renters and investors [8]. Gong (2025) tested Ridge Regression, Decision Tree, Random Forest, and XGBoost on Airbnb data from New York City. After adjusting features and tuning the models, Random Forest had the lowest prediction error and showed that room features, location, and amenities were most important [9]. Fedchenko and Kobets (2025) compared Multiple Linear Regression, Decision Tree, and Random Forest for rental prices in Ukraine and found that machine learning improved both accuracy and transparency [10]. Overall, these studies suggest that tree-based models, especially Random Forest, are effective for rental predictions and are relatively easy to interpret.

Some researchers have tried other ways to predict rental prices. Tan et al. (2024) built a deep learning model that used property features, listing descriptions, and images to predict Airbnb rental prices. They found that combining several types of information improved prediction accuracy compared to models that used only structured data. Highlighting the value of multimodal approaches in housing analytics [11].

Student housing is another key research area since universities affect local rental markets. Jaroensittichai et al. (2024) used machine learning to predict off-campus student housing rents with demographic data, housing features, and renter satisfaction. Their results showed that demand for student housing depends on more than just property features [12]. Still, most of these studies focus on predicting prices for single properties or specific markets, not on rental investment potential across many counties.

Machine learning is also used for other housing problems besides price prediction. Rakhimova et al. (2025) used it to classify tenant reliability using financial history, rental duration, and complaints. Ye et al. (2019) created methods to spot tenants at higher risk of landlord harassment in New York City. These studies show the range of machine learning uses in housing and how it can support private and public decision-making [13].

With machine learning progressing, researchers have investigated ensemble learning and nonlinear models. Abimannan et al. (2023) reviewed ensemble methods and found that combining models usually improves predictions and makes them more robust [14]. He et al. (2025) used spatial machine learning for warehouse rental pricing and found that tree-based models captured complex relationships between location, neighborhood, and market factors better than classic regression [15].

These studies show the growing role of machine learning and data-driven analysis in real estate research. However, much of the existing work focuses on large urban markets or general housing valuation rather than student-oriented rental markets in college towns. This analysis extends previous work by combining housing, demographic, and university enrollment data to evaluate the rental investment potential in college towns across the U.S. The project also emphasizes reproducible data integration and comparative evaluation across multiple data sources to better understand factors associated with rental investment opportunities.

IV. Methodology

A. *Data Collection and Preprocessing*

The first step of this project was to collect and prepare the data before building the machine learning models. Data were collected from four publicly available sources. Zillow provided home values (ZHVI) and rental prices (ZORI), the ACS provided demographic and housing information, IPEDS provided university data, and mortgage rates were obtained from FRED database.

The datasets were merged using county and year identifiers to create a single county-level dataset covering 2012 through 2024. Before training the models, the data were cleaned by correcting inconsistent county identifiers, removing duplicate records, and handling missing values. Missing ACS and IPEDS values were filled using county trends, state-level medians, interpolation, or forward filling, depending on the variable. Counties with insufficient university data were removed, and corrupted Zillow records found during quality checks were corrected.

B. *Exploratory Data Analysis*

Exploratory Data Analysis was performed to better understand the dataset before model development. Summary statistics, distribution plots, correlation analysis, and historical trend charts were used to examine the variables and identify missing values or unusual observations.

The analysis showed that rental yields generally declined between 2012 and 2022 because home values increased faster than rents (Figure 1). Rental yields began to stabilize in 2023 and 2024. Correlation analysis also showed that housing variables, such as median home value and median household income, had a stronger relationship with rental yield than enrollment alone.

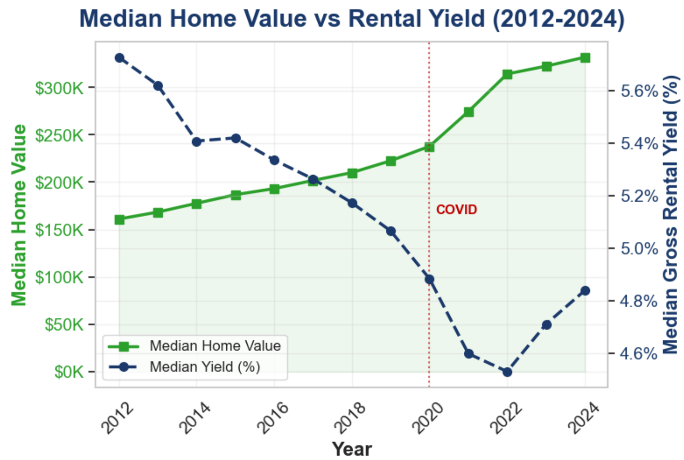

Fig. 1. Median Home Value vs Rental Yield (2012-2024)

C. Feature Engineering

Several new variables were created to improve the predictive performance of the models. These included price-to-income ratio, rent-to-income ratio, renter share, enrollment intensity, student-per-housing-unit ratio, housing deficit, county average yield, previous-year rental yield, and year-over-year growth measures.

During model development, the initial results showed unusually high accuracy. A review of the feature engineering process revealed data leakage caused by variables that included future information. The lag-based variables were rebuilt using only historical data available during each training period, the affected features were removed, and all models were retrained before continuing the analysis.

D. Model Development and Evaluation

Five regression models were evaluated: Ridge Regression, Lasso Regression, Random Forest, LightGBM, and XGBoost. Walk-forward validation was used, with each model trained on earlier years and validated on the following year. A separate 2024 holdout dataset was reserved for the final model evaluation.

Model performance was measured using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), Mean Absolute Percentage Error (MAPE), and the coefficient of determination (R2). Hyperparameter tuning was performed for Random Forest, LightGBM, and XGBoost using randomized search. Weighted and ensemble models were also evaluated to determine whether combining models improved prediction accuracy. Although the ensemble models achieved competitive results, the tuned Random Forest model performed best on the 2024 holdout dataset and was selected as the final model (Table 1).

TABLE I
PLACEHOLDER CAPTION

Model	MAE	R2
Tuned Random Forest	**0.00567**	**0.6581**
Tuned LightGBM	0.00627	0.5870
Tuned XGBoost	0.00605	0.6065
Tree Ensemble	0.00588	0.6314
Full Ensemble	0.00640	0.5855

Figure 2 shows the relationship between the actual and predicted rental yields for the Random Forest model on the 2024 holdout dataset. Most predictions fall close to the perfect prediction line, presenting a good model performance.

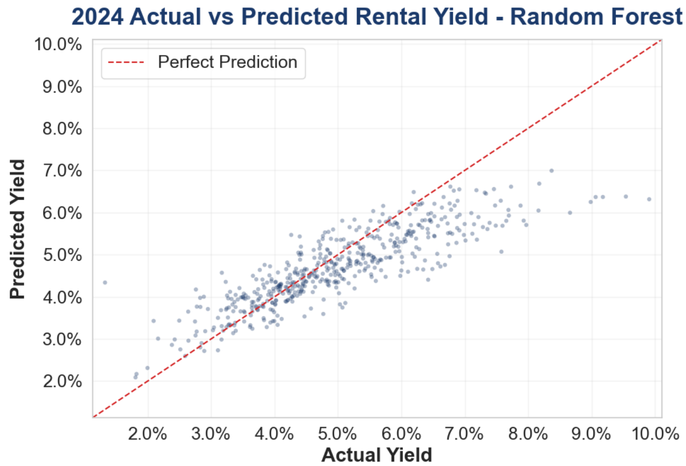

Fig. 2. 2024 Actual vs Predicted Rental Yield - Random Forest

E. Forecasting

The final Random Forest model was retrained using all historical observations from 2012 through 2024 before generating rental yield predictions for 2025 and 2026. The forecasts were then used to compare counties, evaluate historical trends, and identify areas with stronger rental investment potential.

Figure 3 shows the distribution of actual rental yields in 2024 compared with the predicted distributions for 2025 and 2026. The predicted distributions shift toward higher rental yields, with the median increasing from 4.84% in 2024 to 6.10% in 2025 and 6.20% in 2026, suggesting an improvement in rental investment.

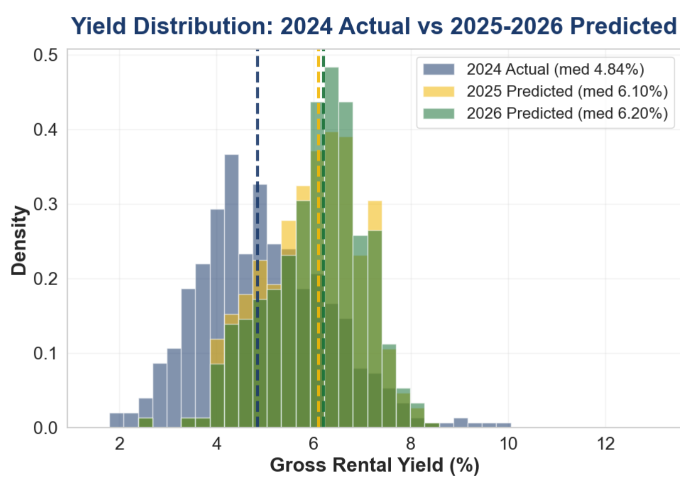

Fig. 3. 2024 Actual vs 2025-2026 Predicted

Figure 4 shows the feature importance results from the final Random Forest model. Median home value was the most important predictor, followed by unemployment rate, median household income, median gross rent, and student housing capacity. This suggests that housing market conditions played a larger role in predicting rental yield than university enrollment alone.

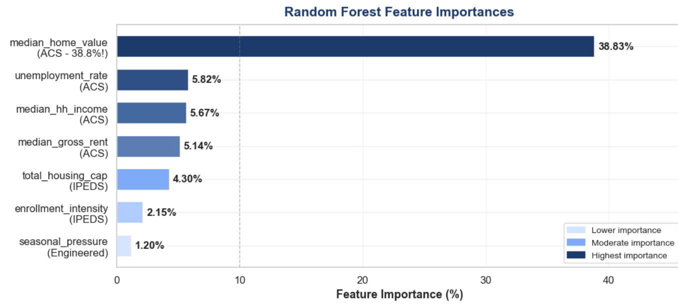

Fig. 4. Random Forest Feature Importance

V. Data Description

The dataset used in this study combines publicly available county-level data on housing, demographics, and the economy, as well as university information, from four sources: Zillow, ACS, IPEDS, and FRED database. Historical data cover the years 2012 through 2024, with additional records generated to forecast 2025 and 2026. After merging the datasets by county and year, the historical dataset contained 35,535 observations across 2,932 counties in 51 states. Applying the county selection criteria resulted in a final modeling dataset of 428 college-town counties. The response variable was gross rental yield, calculated as annual rental income divided by median home value. This metric was selected because it supports rental investment potential to be compared across counties with different housing prices.

A. Data Sources

The project combined information from four public datasets.

- Zillow provided the Zillow Home Value Index (ZHVI) and Zillow Observed Rent Index (ZORI), which were used to calculate historical rental yield.
- ACS provided demographic and housing variables, including population, household income, vacancy rate, educational attainment, and housing characteristics.
- IPEDS provided university information such as enrollment, graduation rate, admissions, and student housing capacity. FRED provided average mortgage rates

B. Feature Set

After preprocessing and feature engineering, the final dataset contained 26 predictor variables. These variables can be grouped into four categories:

- Housing: home value, rent, vacancy rate, renter share, and home value and rent growth.
- Demographic and economic: population, household income, educational attainment, unemployment rate, and mortgage rate.
- University: enrollment, enrollment intensity, housing deficit, student-per-housing-unit ratio, graduation rate, and admissions characteristics.
- Engineered features: price-to-income ratio, rent-to-income ratio, county average yield, previous-year yield, housing deficit growth, enrollment growth, COVID indicator, and other lag-based variables.

C. Data Preparation

The datasets were merged using county and year identifiers before model development. Missing values were handled using county trends, state-level medians, interpolation, or forward filling, depending on the variable. Counties with incomplete university data were removed, and corrupted Zillow records identified during quality checks were corrected.

During model development, a high model accuracy led to a review of the feature engineering process. This review identified data leakage arising from variables that contained future information. The affected variables were rebuilt using only historical data available during each training period, and all models were retrained before continuing the analysis.

Another challenge was reducing potential bias in the data. Some counties did not have complete rental information and required ACS proxy values, while demographic variables were only available through the latest published survey estimates. To reduce these limitations, multiple data sources were combined, walk-forward validation was used during model development, and the final model was evaluated on an independent 2024 holdout dataset.

The final model was deployed through a Streamlit dashboard that allows users to explore county-level results and compare historical and predicted rental yields.

VI. Expected Outcomes

The main objective of this project is to demonstrate how data science methods can be applied to analyze rental investment potential in selected Colorado college towns. The proposed framework consolidates housing market data, demographic information, and university enrollment statistics to identify patterns associated with stronger student rental markets and to examine possible relationships among rental prices, home values, enrollment trends, and local economic conditions. Machine learning models will be used to analyze historical housing and demographic data and evaluate factors that may influence rental investment opportunities.

VII. Timeline

Weeks 1–2: Define practicum topic, project scope. Create an abstract and a problem statement. expected outcomes and timeline. Research college-town housing markets and review related work in real estate analytics and machine learning. Collect housing, demographic, and university enrollment data from Zillow or Kaggle, ACS, and IPEDS, then begin cleaning and organizing the data. Combine housing, demographic, and university datasets into a single analytical dataset.

Week 3: Perform exploratory analysis and create summary statistics, correlations, and initial visualizations.

Week 4: Create investment-related features and evaluate whether additional datasets could improve the analysis.

Week 5: Train baseline machine learning models. Compare model performance and review feature importance results. Troubleshoot and correct data leakage.

Week 6: Tune model parameters for tree based models. Evaluate final models and generate visualizations and summary tables. Create the final presentation.

Week 7: Create an interactive Dashboard.

Week 8: Finalize the final report, the practicum presentation, and the GitHub repository for submission.

VIII. Conclusion

This practicum developed a machine learning model to evaluate rental investment potential across U.S. college-town counties using housing, demographic, economic, and university data from Zillow, ACS, IPEDS, and FRED. A county-level dataset covering 2012 through 2024 was used to generate rental yield forecasts for 2025 and 2026. Several machine learning models were evaluated using walk-forward validation and an independent 2024 holdout dataset. Although LightGBM performed well during cross-validation, the tuned Random Forest model produced the best results on the holdout data and was selected as the final model. The analysis also showed that housing market variables, particularly median home value, were stronger predictors of rental yield than university enrollment alone. A key outcome of this project was finding and fixing data leakage during model development. Rebuilding the affected features and retraining the models led to more reliable results and showed how important careful data preparation and validation are. The final model was also deployed as an interactive dashboard, so users can explore county-level rental yield predictions, compare markets, and see historical and forecast trends.

Interactive Dashboard: https://ml-college-rental-investment.streamlit.app/ Dashboard Demonstration (YouTube): https://www.youtube.com/watch?v=QP2jdGacBAg

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